

Stopping distance

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	What is the stopping distance of a vehicle?	Thinking distance + braking distance	☐
2 ★	What is the thinking distance of a vehicle?	The distance travelled during the driver's reaction time.	☐
3 ★	What is the braking distance of a vehicle?	The distance travelled under the braking force.	☐
4	Name the factor that increases both thinking distance and braking distance.	Speed	☐
5	State the typical range of human reaction times.	0.2 s to 0.9 s	☐
6 ★	Give four factors that affect reaction time.	Tiredness, drugs, alcohol and distractions	☐
7 ★	Give four factors that affect braking distance.	Speed, wet or icy roads, worn tyres and worn brakes	☐
8 ★	Describe the energy transfer when a vehicle brakes.	Energy is transferred from the kinetic store of the vehicle to the thermal store of the brakes.	☐
9	Describe how the braking force needed to stop in a given distance depends on speed.	The greater the speed, the greater the braking force needed.	☐
10 ★	Give two dangers of a very large deceleration.	The brakes may overheat and the driver may lose control.	☐
11	Describe a simple method to measure a person's reaction time.	Drop a ruler between their fingers. The further it falls before they catch it, the longer the reaction time.	☐
12	Explain why braking distance increases with speed.	The vehicle has more kinetic energy, so the brakes must do more work to stop it.	☐

QUESTION		ANSWER	✓
13	The speed of a vehicle doubles. State what happens to the thinking distance.	It doubles (it is directly proportional to speed).	☐
14	Which does speed affect more: thinking distance or braking distance?	Braking distance	☐